

ECE 443 – Introduction to Computer Cyber Security
ECE 518 – Computer Cyber Security
Fall 2026

Instructor: Professor Jia Wang

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Prerequisites: Computer programming; digital logic and computer organization; probability.

Class Time and Location: Mon./Wed. 11:25 AM – 12:40 PM, Robert A. Pritzker Science Ctr 111

Class Home Page: <https://wngjia.github.io/ece443-web/>

Required Textbook:

- [UC] “Understanding Cryptography: A Textbook for Students and Practitioners”
C. Paar and J. Pelzl, Springer, 2010. ISBN-13: 978-3642446498. Download at
<https://link-springer-com.ezproxy.gl.iit.edu/book/10.1007/978-3-642-04101-3>

Recommended Textbook:

- [ICS] “Introduction to Computer Security” M. Bishop, Addison-Wesley, 2005. ISBN: 0321247442

Computer Requirement:

- A computer supporting development in Go
- Internet access to common code and package repositories like GitHub

Course Descriptions:

ECE 443: This course gives students a clear understanding of computer and cyber security as threats and defense mechanisms backed by mathematical and algorithmic guarantees. Key topics covered include introductory number theory and complexity theory, cryptography and applications, system security, digital forensics, software and hardware security, and side-channel attacks. Course projects will survey trending topics and provide hands-on experiences on languages, libraries, and tools supporting state-of-the-art cryptography systems.

ECE 518: This course builds on the foundations of computer and cyber security through a rigorous treatment of threat models and the mathematical and algorithmic guarantees that underlie modern security mechanisms. Key topics covered include introductory number theory and complexity theory, cryptography and applications, cryptocurrency and secure multi-party computation, system security, digital forensics, software and hardware security, and side-channel attacks. Course projects will survey trending topics and provide hands-on experiences on languages, libraries, and tools supporting state-of-the-art cryptography systems and advanced applications.

Grading:

- 4 Homeworks: 5 points each for a total of 20 points.
- 6 Projects: 15 points each for a total of 90 points.
- Midterm Exam: 30 points.
- ECE 443: $A \geq 90$ / $B \geq 80$ / $C \geq 60$ / D (undergraduate only) ≥ 55 .
- ECE 518: $A \geq 115$ / $B \geq 100$ / $C \geq 80$.

Homework and Project Policy: Late homeworks and project reports will not be graded. Discussions on homeworks and projects are encouraged, but copying will call for disciplinary action.

Exam Policy: Close book, close note, cheat sheet allowed. Makeup exams will NOT be given.

Lecture Schedule (tentative):

No.	Date	Topic	Chapters	HW Out	Project Due
1, 2	8/17, 8/19	Introduction	ICS 1, UC 1		
3, 4	8/24, 8/26	Stream and Block Ciphers	UC 2-5	HW #1	
5, 6	8/31, 9/2	Go, Cryptographic Hash Function	UC 11		
7	9/7, 9/9	Authenticated Encryption	UC 12		PRJ #1
8, 9	9/14, 9/16	Complexity Theory, Number Theory, RSA	UC 6, 7	HW #2	
10,11	9/21, 9/23	Diffie-Hellman, Digital Signatures	UC 8, 10		PRJ #2
12,13	9/28, 9/30	Key Establishment	UC 13		
14,15	10/5, 10/7	Password Security, OpenSSL			
16	10/12 , 10/14	Midterm Exam			
17,18	10/19, 10/21	Consensus and Cryptocurrency			
19,20	10/26, 10/28	Smart Contract, Bitcoin Security			PRJ #3
21,22	11/2, 11/4	Secure Multi-Party Computation		HW #3	
23,24	11/9, 11/11	Access Control	ICS 2-7, 14		
25,26	11/16, 11/18	Digital Forensics		HW #4	
27	11/23, 11/25	Malware	ICS 19		PRJ #4
28,29	11/30, 12/2	Hardware Security, Side-Channel Attacks			PRJ #5
	12/7-12/11	No Final Exam			PRJ #6

ECE 443 Course Objectives (ABET)

After completing this course, you should be able to:

1. Describe computer cyber security as threats and defense mechanisms.
2. Understand stream ciphers, block ciphers, cryptographic hash functions, and public-key cryptography.
3. Explain authenticated encryption, man-in-the-middle attack, perfect forward secrecy, and their impact on secure communication protocol designs.
4. Understand system security concepts including security policies and access control.
5. Describe vulnerabilities in software and hardware systems.
6. Explain digital forensics processes.

AI Usage Policy: The instructor may use AI tools to develop course content and please be assured that the instructor maintains final oversight, review, and ultimate responsibility for all course materials.

You are encouraged to use AI tools as part of learning and project development in this course with proper acknowledgment. Nevertheless, you are ultimately responsible for the content you submit, regardless of its source.

ADA Statement: Reasonable accommodations will be made for students with documented disabilities. In order to receive accommodations, students must obtain a letter of accommodation from the Center for Disability Resources and make an appointment to speak with me as soon as possible. The Center for Disability Resources is located in the Life Sciences Building, room 218, 312-567-5744 or disabilities@iit.edu.

Sexual Harassment and Discrimination Information: Illinois Tech prohibits sexual harassment by any member of the community. This includes harassment among students, staff, or faculty. Sexual harassment by a faculty member or teaching assistant of a student over whom they have authority can be particularly serious. Such conduct may easily create an intimidating, hostile, or offensive environment. Accordingly, Illinois Tech strives to provide for its students, faculty and staff an educational and employment environment free of sexual harassment.

Illinois Tech encourages anyone experiencing sexual harassment to speak with the Office of Title IX Compliance for information on support options and the resolution process. You can report sexual harassment electronically at iit.edu/incidentreport or contact either the Title IX Coordinator, Virginia Foster at foster@illinoistech.edu, the Title IX Specialist, Ryley Andrews at randrews2@illinoistech.edu.

To speak with someone confidentially, contact Illinois Tech's Confidential Advisor at (773) 275 8340 or a licensed practitioner in Illinois Tech's Student Health and Wellness Center by emailing student.health@illinoistech.edu or calling (312)567-7550. For a comprehensive list of resources, visit the Office of Title IX Compliance website at <https://www.iit.edu/title-ix/resources>.

University Resources for Students' Rights, Responsibilities, and Related Policies:

- The Center for Disability Resources (CDR) Rights and Responsibilities
<https://www.iit.edu/cdr/faqs/rights-and-responsibilities>
- The Family Educational Rights and Privacy Act (FERPA)
<https://www.iit.edu/registrar/students-and-alumni/ferpa>
- Code of Conduct (from Student Handbook)
<https://www.iit.edu/student-affairs/student-handbook/fine-print/code-conduct>
- Policies, Procedures, and Regulations (from Student Handbook)
<https://www.iit.edu/student-affairs/student-handbook/fine-print/policies-regulations-and-procedures>